



VP AIR TEST NEET (2025-26)

PART TEST - 09

DROPPER

DURATION: 180 Minutes

Date: 01-02-2026

M. MARKS: 720

ANSWER KEY

1. (2)	46. (1)	91. (2)	136. (2)
2. (4)	47. (2)	92. (3)	137. (3)
3. (2)	48. (4)	93. (2)	138. (2)
4. (2)	49. (3)	94. (2)	139. (1)
5. (2)	50. (2)	95. (3)	140. (3)
6. (1)	51. (3)	96. (4)	141. (4)
7. (2)	52. (3)	97. (2)	142. (1)
8. (1)	53. (4)	98. (1)	143. (1)
9. (1)	54. (3)	99. (3)	144. (2)
10. (3)	55. (3)	100. (2)	145. (1)
11. (1)	56. (1)	101. (4)	146. (4)
12. (2)	57. (3)	102. (1)	147. (2)
13. (2)	58. (3)	103. (3)	148. (3)
14. (2)	59. (1)	104. (3)	149. (3)
15. (1)	60. (2)	105. (3)	150. (2)
16. (1)	61. (3)	106. (2)	151. (3)
17. (2)	62. (2)	107. (1)	152. (2)
18. (4)	63. (1)	108. (1)	153. (2)
19. (2)	64. (1)	109. (2)	154. (2)
20. (1)	65. (1)	110. (3)	155. (3)
21. (1)	66. (3)	111. (1)	156. (2)
22. (2)	67. (4)	112. (3)	157. (1)
23. (1)	68. (4)	113. (1)	158. (4)
24. (1)	69. (1)	114. (4)	159. (2)
25. (2)	70. (1)	115. (1)	160. (3)
26. (1)	71. (3)	116. (2)	161. (2)
27. (1)	72. (1)	117. (3)	162. (4)
28. (3)	73. (2)	118. (2)	163. (3)
29. (4)	74. (4)	119. (3)	164. (3)
30. (2)	75. (3)	120. (1)	165. (2)
31. (2)	76. (1)	121. (3)	166. (3)
32. (1)	77. (4)	122. (4)	167. (4)
33. (2)	78. (3)	123. (3)	168. (4)
34. (1)	79. (3)	124. (4)	169. (1)
35. (3)	80. (1)	125. (1)	170. (2)
36. (2)	81. (1)	126. (3)	171. (1)
37. (4)	82. (2)	127. (2)	172. (4)
38. (4)	83. (2)	128. (3)	173. (3)
39. (3)	84. (1)	129. (1)	174. (3)
40. (2)	85. (4)	130. (2)	175. (2)
41. (1)	86. (1)	131. (4)	176. (3)
42. (3)	87. (4)	132. (2)	177. (4)
43. (3)	88. (1)	133. (3)	178. (4)
44. (3)	89. (1)	134. (4)	179. (1)
45. (4)	90. (2)	135. (1)	180. (1)

HINTS & SOLUTIONS

1. (2)

$$\text{Induced EMF } |e| = \left| \frac{d\phi}{dt} \right|$$

$$\frac{d}{dt}(5t^2 + 3t + 16) = 10t + 3$$

$$\text{At } t = 4: |e| = 10(4) + 3 = 43 \text{ V}$$

(NCERT Ed. 2025-26, 12th, Page No. 158)

2. (4)

Peak Electric Field

$$E_0 = B_0 c = (2 \times 10^{-7})(3 \times 10^8) = 60 \text{ V/m}$$

The wave equation contains $(kx + \omega t)$, implying propagation in the $-x$ direction $(-\hat{i})$.

$$\text{Direction of propagation is } \hat{k}_{prop} = \hat{E} \times \hat{B}.$$

We have: $-\hat{i} = \hat{E} \times \hat{j}$ the Electric field must be along $\hat{k}$ (z-axis).

(NCERT Ed. 2025-26, 12th, Page No. 208)

3. (2)

$$\omega = 100 \text{ rad/s.}$$

$$\text{Inductive Reactance } X_L = \omega L = 100 \times 0.4 = 40 \Omega$$

Capacitive Reactance

$$X_C = \frac{1}{\omega C} = \frac{1}{100 \times 100 \times 10^{-6}} = \frac{1}{10^{-2}} = 100 \Omega$$

$$\text{Net Reactance } X = |X_L - X_C| = |40 - 100| = 60 \Omega$$

$$\text{Impedance } Z = \sqrt{R^2 + X^2} = \sqrt{80^2 + 60^2}$$

$$= \sqrt{6400 + 3600} = \sqrt{10000} = 100 \Omega$$

(NCERT Ed. 2025-26, 12th, Page No. 187)

4. (2)

$$|e| = L \frac{di}{dt} = \frac{\mu_0 N^2 A}{l} \times \frac{di}{dt}$$

$$= \frac{4\pi \times 10^{-7} \times (40)^2 \times 12 \times 10^{-4}}{4\pi \times 10^{-2}} \times 1$$

$$= 1.92 \times 10^{-5} \text{ V}$$

(NCERT Ed. 2025-26, 12th, Page No. 168)

5. (2)

The total charge flow depends only on the change in flux and resistance, not on time or inductance.

$$Q = \frac{|\Delta\phi|}{R}$$

$$\Delta\phi = 25 - 5 = 20 \text{ Wb}$$

$$Q = \frac{20}{10} = 2 \text{ C}$$

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6. (1)

At $t = 0$, the inductor opposes current change infinitely (Open Circuit). Current in Branch 1 $(L + R_1) = 0$.

The circuit effectively consists of the Battery, Internal Resistance ($r = 2\Omega$), and R_2 (2Ω).

$$\text{Total Resistance} = R_2 + r = 2 + 2 = 4\Omega.$$

$$\text{Current } I = \frac{12}{4} = 3 \text{ A}$$

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7. (2)

$$V_p I_p \times \eta = V_s I_s$$

$$220 \times I_p \times 0.8 = 22 \times 2$$

$$176 I_p = 44$$

$$I_p = \frac{44}{176} = 0.25 \text{ A}$$

(NCERT Ed. 2025-26, 12th, Page No. 195)

8. (1)

$$I_0 = 0.1 \text{ A}$$

$$V_{rms} = \frac{100}{\sqrt{2}},$$

$$I_{rms} = \frac{0.1}{\sqrt{2}}$$

$$\text{Phase } \phi = 60^\circ. \text{ Power factor } \cos 60^\circ = 0.5$$

$$P = V_{rms} I_{rms} \cos \phi = \frac{100 \times 0.1}{2} \times 0.5$$

$$= \frac{10}{2} \times 0.5 = 2.5 \text{ W}$$

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9. (1)

$$|e| = M \left| \frac{dI}{dt} \right| = M \cdot 10 \cdot 100\pi \cos(100\pi t)$$

$$e_{max} = 0.05 \times 1000\pi = 50\pi \text{ V}$$

(NCERT Ed. 2025-26, 12th, Page No. 167)

10. (3)

$$e = \frac{d\phi}{dt} = B \cdot \frac{d(\pi r^2)}{dt} = B \cdot (2\pi r) \cdot \frac{dr}{dt}$$

$$\text{Given: } e = 4\pi \times 10^{-6} \text{ V}, B = 0.04 \text{ T}$$

$$r = 5 \text{ cm} = 5 \times 10^{-2} \text{ m}$$

$$4\pi \times 10^{-6} = 0.04 \times (2\pi \times 5 \times 10^{-2}) \times \frac{dr}{dt}$$

$$4\pi \times 10^{-6} = 4\pi \times 10^{-3} \times \frac{dr}{dt}$$

$$\frac{dr}{dt} = \frac{4\pi \times 10^{-6}}{4\pi \times 10^{-3}} = 10^{-3} \text{ m/s}$$

$$\frac{dr}{dt} = 1 \text{ mm/s}$$

(NCERT Ed. 2025-26, 12th, Page No. 158)

11. (1)
Displacement current inside the capacitor equals the conduction current in the connecting wires.
For a discharging RC circuit, $i(t) = \frac{V}{R} e^{-t/RC}$
(NCERT Ed. 2025-26, 12th, Page No. 204)

12. (2)
We know, eddy currents are produced in the form of whirls.
 $\therefore$ In figure B, the length for flow of current is maximum.
Thus, it will have maximum resistance and hence minimum current.
(NCERT Ed. 2025-26, 12th, Page No. 196)

13. (2)
 $M = \frac{\mu_0 N_1 N_2 A}{L}$
 $M = \frac{(4\pi \times 10^{-7}) \times (4000) \times (100) \times (10^{-3})}{0.5}$
 $M = \frac{4\pi \times 10^{-7} \times 4 \times 10^5 \times 10^{-3}}{0.5}$
 $M = \frac{16\pi \times 10^{-5}}{0.5}$
 $M = 32\pi \times 10^{-5} \text{ H}$
(NCERT Ed. 2025-26, 12th, Page No. 166)

14. (2)
 $\varepsilon = Bvl$ Where l is the length of the edge perpendicular to the velocity. Since B and v are the same for all loops, $\varepsilon \propto l$ (vertical height).
Length of Loop 4 = $3L$
Length of Loop 3 = $2L$
Length of Loops 1 & 2 = L
Therefore, the order is $4 > 3 > 2 = 1$.
 $\varepsilon_4 > \varepsilon_3 > \varepsilon_2 = \varepsilon_1$
(NCERT Ed. 2025-26, 12th, Page No. 162)

15. (1)
Time constant $\tau = L / R = 1 \text{ s}$
 $I = I_{\max} (1 - e^{-t/\tau})$
For $I = 0.5 I_{\max}$, $e^{-t/\tau} = 0.5 \Rightarrow t = \tau \ln 2$
 $t = 1 \times 0.693 = 0.693 \text{ s}$
(NCERT Ed. 2025-26, 12th, Page No. 169)

16. (1)
 $L = \mu_0 n^2 Al$
($n = N / l = 1000$)
 $L = 4\pi \times 10^{-7} \times (10^6) \times 10^{-3} \times 1 = 4\pi \times 10^{-4} \text{ H}$
Energy $U = \frac{1}{2} LI^2 = \frac{1}{2} (4\pi \times 10^{-4}) (100)^2$
 $U = 2\pi \times 10^{-2} \text{ J}$
(NCERT Ed. 2025-26, 12th, Page No. 169)

17. (2)
Power $P = Fv = (IlB)v = \left(\frac{Blv}{R} lB \right) v = \frac{B^2 l^2 v^2}{R}$
 $P \propto v^2$
(NCERT Ed. 2025-26, 12th, Page No. 160)

18. (4)
 $|e| = L \frac{\Delta I}{\Delta t} = 5 \times 10^{-3} \times \frac{3}{0.05} = 5 \times 10^{-3} \times 60$
 $= 300 \times 10^{-3} = 0.3 \text{ V}$
(NCERT Ed. 2025-26, 12th, Page No. 168)

19. (2)
A choke coil (pure inductor) has
 $P_{\text{avg}} = VI \cos(90^\circ) = 0$
It restricts current via reactance without energy loss.
(NCERT Ed. 2025-26, 12th, Page No. 191)

20. (1)
 $\omega = 2\pi f$
 $e = \frac{1}{2} B \omega a^2 = \frac{1}{2} B (2\pi f) a^2 = \pi B a^2 f$
(NCERT Ed. 2025-26, 12th, Page No. 164)

21. (1)
 $\tan \phi = \frac{X_L}{R}$
Initially $\phi = 45^\circ \Rightarrow X_L = R$
If frequency doubles, $X_L' = 2X_L$. R stays same.
 $\tan \phi' = \frac{2X_L}{R} = 2(1) = 2$
 $\phi' = \tan^{-1}(2)$
(NCERT Ed. 2025-26, 12th, Page No. 191)

22. (2)
 $LC = \text{constant}$
 $L_{\text{new}} (4C) = L_{\text{old}} C$
 $L_{\text{new}} = L_{\text{old}} / 4$
(NCERT Ed. 2025-26, 12th, Page No. 189)

23. (1)
In Situation A: The coils are coaxial and their planes are parallel. The magnetic field lines from the larger coil pass perpendicularly through the smaller coil, resulting in maximum flux linkage. Since $M \propto \text{Flux Linkage}$, mutual inductance is maximum in situation (A).
(NCERT Ed. 2025-26, 12th, Page No. 167)

24. (1)
Magnitude $B = E / c = 6.3 / (3 \times 10^8) = 2.1 \times 10^{-8} \text{ T}$.
Direction: Wave ($\hat{i}$), E ($\hat{j}$).
 $\hat{k}_{prop} = \hat{E} \times \hat{B} \Rightarrow \hat{i} = \hat{j} \times \hat{B}$. $\hat{B}$ must be $\hat{k}$ because
 $\hat{j} \times \hat{k} = \hat{i}$.
(NCERT Ed. 2025-26, 12th, Page No. 208)

25. (2)
 $V_{rms} = 200 \text{ V}$
 $X_C = \frac{1}{\omega C} = \frac{1}{100 \times 100 \times 10^{-6}} = \frac{1}{10^{-2}} = 100 \Omega$
 $I_{rms} = \frac{V_{rms}}{X_C} = \frac{200}{100} = 2 \text{ A}$
(NCERT Ed. 2025-26, 12th, Page No. 185)

26. (1)
At resonance $X_L = X_C$, so $Z = R$ (Minimum).
Since $I = V / Z$, current is maximum.
(NCERT Ed. 2025-26, 12th, Page No. 189)

27. (1)
Statement I is the correct Maxwell-Ampere law.
Statement II is incorrect; displacement current
 $I_d = \epsilon_0 \frac{d\phi_E}{dt}$ is produced only when the electric
field is changing (time-varying).
(NCERT Ed. 2025-26, 12th, Page No. 204)

28. (3)
Velocity (v): $\frac{\text{Distance}}{\text{Time}} = \frac{1 \text{ m}}{2 \text{ s}} = 0.5 \text{ m/s}$
Induced EMF (ϵ): $Bvl = 2 \times 0.5 \times 1 = 1 \text{ V}$
Power (P): $\frac{\epsilon^2}{R} = \frac{1^2}{5} = 0.2 \text{ W}$
Work (W): $P \times t = 0.2 \text{ W} \times 2 \text{ s} = 0.4 \text{ J}$
(NCERT Ed. 2025-26, 12th, Page No. 160)

29. (4)
Because the phase difference is clearly less than 90° (current is not at its minimum when voltage is zero), it is an $L-R$ circuit rather than a pure inductor.
(NCERT Ed. 2025-26, 12th, Page No. 183)

30. (2)
In an EM wave, $\hat{E} \times \hat{B} \parallel \hat{v}$ (direction of propagation).
 $\hat{k} \times \hat{B} = \frac{1}{\sqrt{2}}(\hat{i} + \hat{j})$
Using cross-product rules: $\hat{k} \times \hat{i} = \hat{j}$ and
 $\hat{k} \times (-\hat{j}) = \hat{i}$
So, $\hat{B} = \frac{1}{\sqrt{2}}(\hat{j} - \hat{i})$
(NCERT Ed. 2025-26, 12th, Page No. 208)

31. (2)
Speed in medium
 $v = \frac{c}{\sqrt{\mu_r \epsilon_r}} = \frac{c}{\sqrt{1 \times 9}} = \frac{c}{3} = 10^8 \text{ m/s}$
Relationship: $B_0 = \frac{E_0}{v} = \frac{12}{10^8} = 1.2 \times 10^{-7} \text{ T}$
(NCERT Ed. 2025-26, 12th, Page No. 207)

32. (1)
Total energy density $u = u_E + u_B$
Since $u_E = u_B$ in an EM wave,
 $u_{avg} = 2 \times (\text{avg } u_B) = 2 \times \frac{B_0^2}{4\mu_0} = \frac{B_0^2}{2\mu_0}$.
(NCERT Ed. 2025-26, 12th, Page No. 214)

33. (2)
 $\omega = 6 \times 10^8 \text{ rad/s}$.
Velocity $v = \frac{c}{n} = \frac{3 \times 10^8}{2} = 1.5 \times 10^8 \text{ m/s}$.
 $k = \frac{\omega}{v} = \frac{6 \times 10^8}{1.5 \times 10^8} = 4 \text{ m}^{-1}$.
(NCERT Ed. 2025-26, 12th, Page No. 207)

34. (1)
Ultraviolet (UV) Rays: Due to their high energy and ability to be focused into narrow beams, they are used in LASIK to reshape the cornea.
Microwaves: Their short wavelengths (sub-centimeter to millimeter) allow for minimal diffraction, making them ideal for Radar and aircraft navigation.
Infrared (IR) Radiation: Known as heat waves, they provide deep tissue heating used in physiotherapy to treat muscle strains.
X-rays: Their high penetrating power through soft tissue but absorption by dense matter makes them the standard for detecting bone fractures.
(NCERT Ed. 2025-26, 12th, Page No. 211)

35. (3)
 $Z = X_L - X_C = 10 \Omega$
 $I_0 = \frac{V_0}{Z} = \frac{20}{10} = 2 \text{ A}$
 $I_{rms} = \frac{2}{\sqrt{2}} = \sqrt{2} \text{ A}$
(NCERT Ed. 2025-26, 12th, Page No. 187)

36. (2)
 $\lambda = \frac{c}{f} = \frac{3 \times 10^8}{3 \times 10^{10}} = 0.01 \text{ m}$
 $B_0 = \frac{E_0}{c} = \frac{96}{3 \times 10^8} = 32 \times 10^{-8} \text{ T}$
Electric field and magnetic field has zero phase difference.
(NCERT Ed. 2025-26, 12th, Page No. 214)

37. (4)

$$V = \frac{10}{8.85} \times 10^{-6} \text{ m}^3$$

$$U = \frac{1}{2} \epsilon_0 E_0^2 V$$

$$= \frac{1}{2} \times (8.85 \times 10^{-12}) \times (60)^2 \times \frac{10^{-5}}{8.85} = 1.8 \times 10^{-14} \text{ J}$$

(NCERT Ed. 2025-26, 12th, Page No. 214)

38. (4)

$$e = -L \frac{di}{dt}$$

For $0 \leq t \leq \frac{T}{4}$, $\frac{di}{dt}$ = negative and constant

e = positive and constant

for $\frac{T}{4} \leq t \leq \frac{T}{2}$, $\frac{di}{dt} = 0$

$\Rightarrow e = 0$

similarly,

for $\frac{T}{2} \leq t \leq \frac{3T}{4}$

$\Rightarrow e$ = negative and constant

for $\frac{3T}{4} \leq t \leq T$, $\frac{di}{dt} = 0$

$\Rightarrow e = 0$

(NCERT Ed. 2025-26, 12th, Page No. 193)

39. (3)

The effective current is simply the given current,

$$I_{DC} = 10 \text{ A}$$

The effective (RMS) current is the peak current divided by the square root of 2

$$I_{rms} = 10 / \sqrt{2} \text{ A}$$

$$\text{Ratio} = (10)^2 : (10 / \sqrt{2})^2$$

$$\text{Ratio} = 2 : 1$$

(NCERT Ed. 2025-26, 12th, Page No. 180)

40. (2)

$$X_L = \omega \times L = 100 \text{ rad/s} \times 0.4 \text{ H} = 40 \Omega$$

$$Z = \sqrt{R^2 + X_L^2} = \sqrt{(30^2 + 40^2)} = \sqrt{(900 + 1600)}$$

$$= \sqrt{2500} = 50 \Omega$$

$$I_{rms} = V_{rms} / Z = \frac{100 \text{ V}}{50 \Omega} = 2 \text{ A}$$

The wattless current is $I_{rms} \times \sin(\phi)$

$$\sin(\phi) = X_L / Z = \frac{40}{50} = 0.8$$

$$\text{Wattless Current} = 2 \text{ A} \times 0.8 = 1.6 \text{ A}$$

(NCERT Ed. 2025-26, 12th, Page No. 192)

41. (1)

For $0 \leq t \leq T/2$, the line is $i(t) = (2i_0 / T) \times t$

For $T/2 \leq t \leq T$, the line is

$$i(t) = (2i_0 / T) \times (t - T)$$

$$I_{rms}^2 = \left(\frac{1}{T} \right) \left[\int_0^{T/2} \left(\frac{2i_0}{T} \right)^2 t^2 dt + \int_{T/2}^T \left(\frac{2i_0}{T} \right)^2 (t - T)^2 dt \right]$$

Due to the symmetry of the squared waveform, the value of both integrals is the same.

Solving the first integral:

$$\int_0^{T/2} \left(\frac{4i_0^2}{T^2} \right) t^2 dt = \left(\frac{4i_0^2}{T^2} \right) \left[\frac{t^3}{3} \right]_0^{T/2}$$

$$= \left(\frac{4i_0^2}{T^2} \right) \times \left(\frac{T^3}{24} \right) = \frac{i_0^2 T}{6}$$

$$\text{So, the total integral is } \left(\frac{i_0^2 T}{6} \right) + \left(\frac{i_0^2 T}{6} \right) = \frac{i_0^2 T}{3}$$

$$I_{rms}^2 = \left(\frac{1}{T} \right) \times \left(\frac{i_0^2 T}{3} \right) = \frac{i_0^2}{3}$$

$$I_{rms} = \sqrt{\frac{i_0^2}{3}} = \frac{i_0}{\sqrt{3}}$$

(NCERT Ed. 2025-26, 12th, Page No. 179)

42. (3)

Displacement current will be equal to

$$\epsilon_0 \frac{d\phi_E}{dt} = \epsilon_0 \times \frac{6t}{\epsilon_0} = 6t \text{ where } t \text{ is in second.}$$

$$= 30 \text{ A}$$

(NCERT Ed. 2025-26, 12th, Page No. 203)

43. (3)

Since the current leads the voltage by $\frac{\pi}{4}$ (an angle

between 0 and $\frac{\pi}{2}$), it indicates a circuit with both resistance and capacitance. A purely capacitive circuit would have a lead of exactly $\frac{\pi}{2}$.

Thus, the unknown element must be a combination of a resistor and a capacitor. So, it can either be RC in series or LCR in series in the box.

(NCERT Ed. 2025-26, 12th, Page No. 191)

44. (3)

Eddy currents are induced currents in the bulk of the core material. They cause energy loss. To reduce them, the core is made of thin laminated sheets of soft iron, insulated from each other. This increases the resistance path for the eddy currents, thereby reducing their magnitude and the associated losses.

(NCERT Ed. 2025-26, 12th, Page No. 196)

45. (4)

$$\text{Emf through circular loop} = -\frac{d\phi_B}{dt} = -B \frac{dA}{dt}$$

where A is area of loop

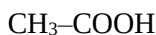
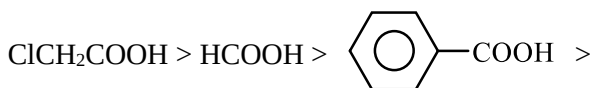
$$= -B \times 2\pi r \frac{dr}{dt} = -0.025 \times 2\pi \times 2 \times 10^{-2} \times 3 \times 10^{-3} \text{ volt}$$

$$= 3\pi \mu \text{ V}$$

(NCERT Ed. 2025-26, 12th, Page No. 158)

46. (1)

Acidic nature:

(NCERT Ed. 2025-26 12th Part-II Page No. 251)

47. (2)

Hinsberg reagent: Benzene sulphonyl chloride:

(NCERT Ed. 2025-26 12th Part-II Page No. 271)

48. (4)

Glycogen is a polysaccharide as it yields a large number of monosaccharide units on hydrolysis.

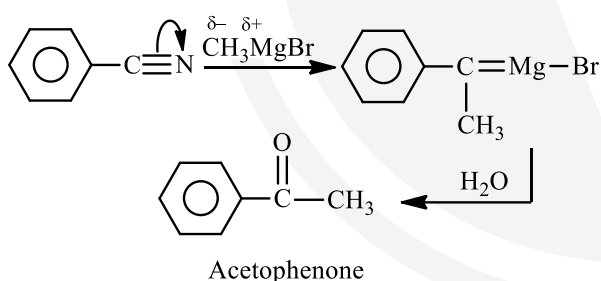
(NCERT Ed. 2025-26 12th Part-II Page No. 282)

49. (3)

- Thin layer chromatography is based on the principle of differential adsorption.
- Partition chromatography is based on continuous differential partitioning of components of a mixture between stationary and mobile phases.

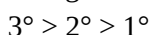
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50. (2)

(NCERT Ed. 2025-26 12th Part-II Page No. 234)

51. (3)

In vapour phase for aliphatic amines order of basic strength:

(NCERT Ed. 2025-26 12th Part-II Page No. 268)

52. (3)

Thyroxine is an iodinated derivative of amino acid tyrosine.

(NCERT Ed. 2025-26 12th Part-II Page No. 300)

53. (4)

Commonly used adsorbent in column chromatography is alumina.

(NCERT Ed. 2025-26 11th Part-II Page No. 283)

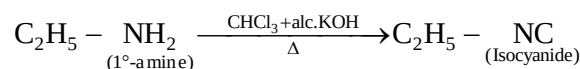
54. (3)

- Camphor is well known for its odour.
- Higher fatty acids are used for manufacture of soaps and detergents.

(NCERT Ed. 2025-26 12th Part-II Page No. 254)

55. (3)

Carbylamine reaction:

(NCERT Ed. 2025-26 12th Part-II Page No. 271)

56. (1)

DNA contains four bases:

Adenine, Guanine, Cytosine and Thymine.

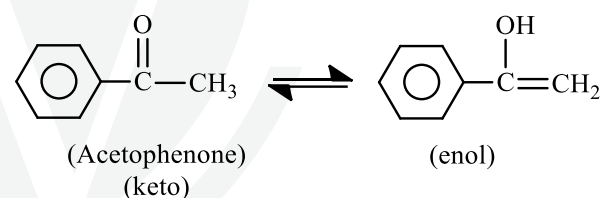
(NCERT Ed. 2025-26 12th Part-II Page No. 298)

57. (3)

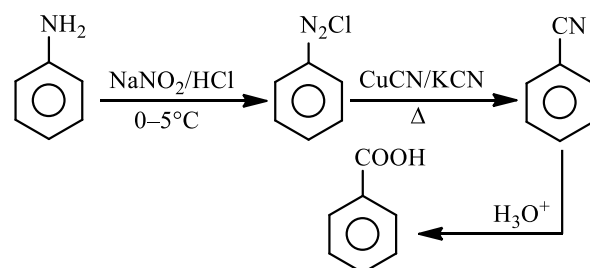
$[\text{Fe}(\text{SCN})]^{2+}$	Red
$[\text{Fe}(\text{CN})_5\text{NOS}]^{4-}$	Violet
$\text{Fe}_4[\text{Fe}(\text{CN})_6]_3 \cdot x\text{H}_2\text{O}$	Prussian blue
PbS	Black

(NCERT Ed. 2025-26 11th Part-II Page No. 285)

58. (3)

(NCERT Ed. 2025-26 12th Part-II Page No. 241)

59. (1)

(NCERT Ed. 2025-26 12th Part-II Page No. 275)

60. (2)

Vitamin C	Scurvy
Vitamin A	Xerophthalmia
Vitamin B ₆	Convulsions
Vitamin E	Increased fragility of RBCs

(NCERT Ed. 2025-26 12th Part-II Page No. 296)

61. (3)
- Fractional distillation method is used to separate different fractions of crude oil in petroleum industry.
 - Simple distillation method is used to separate the liquids having sufficient difference in their boiling points.
- (NCERT Ed. 2025-26 11th Part-II Page No. 279)

62. (2)
- Aldol condensation reaction is given by carbonyl compounds containing α -H atom.
- (NCERT Ed. 2025-26 12th Part-II Page No. 241)

63. (1)
- Coupling reaction is an example of Electrophilic substitution reaction.
 - Amines are said to be 'simple' when all the alkyl or aryl groups are the same and 'mixed' when they are different.
- (NCERT Ed. 2025-26 12th Part-II Page No. 260)

64. (1)
- Sugar present in milk is known as lactose.
- (NCERT Ed. 2025-26 12th Part-II Page No. 288)

65. (1)
- Compound containing both N and S will give blood red colour during lassaingne's test for nitrogen.
- (NCERT Ed. 2025-26 11th Part-II Page No. 285)

66. (3)
- Compounds containing CH_3CO group or $\text{CH}_3\text{CH}(\text{OH})$ group will give iodoform on reaction with I_2/NaOH .
- (NCERT Ed. 2025-26 12th Part-II Page No. 240)

67. (4)
-
- (NCERT Ed. 2025-26 12th Part-II Page No. 262)

68. (4)
-
- Glucose
- Gluconic acid
- (NCERT Ed. 2025-26 12th Part-II Page No. 283)

69. (1)
- Leibig's method is used for the estimation of carbon and hydrogen.
- (NCERT Ed. 2025-26 11th Part-II Page No. 284)

70. (1)
- $$\text{R}-\text{CN} \xrightarrow[\text{(ii) H}_2\text{O}]{\text{(i) AlH(i-Bu)}_2} \text{R}-\text{CHO}$$
- Aldehyde
- (NCERT Ed. 2025-26 12th Part-II Page No. 232)

71. (3)
-
- (NCERT Ed. 2025-26 12th Part-II Page No. 263)

72. (1)
- Nucleotides are joined together by phosphodiester linkage between 5' and 3' carbon atoms of the pentose sugar.
- (NCERT Ed. 2025-26 12th Part-II Page No. 298)

73. (2)
- $$\% \text{ of P} = \frac{62 \times 0.111 \times 100}{222 \times 0.2} = 15.5\%$$
- (NCERT Ed. 2025-26 11th Part-II Page No. 290)

74. (4)
-
- (Ethanoic anhydride)
- (NCERT Ed. 2025-26 12th Part-II Page No. 251)

75. (3)
-
- (NCERT Ed. 2025-26 12th Part-II Page No. 269)

76. (1)
- During denaturation of proteins they lose their biological activity because globules unfold and helix get uncoiled.
- (NCERT Ed. 2025-26 12th Part-II Page No. 294)

77. (4)
- Camphor sublimes on heating while NaCl does not sublime on heating.
- (NCERT Ed. 2025-26 11th Part-II Page No. 279)

78. (3)
-
- Propanone
- (NCERT Ed. 2025-26 12th Part-II Page No. 231)

79. (3)
-
- (Sulphanilic acid)
- (NCERT Ed. 2025-26 12th Part-II Page No. 273)

80. (1)
- Two cyclic hemiacetal forms of glucose differ only in the configuration of hydroxyl group at C1 are called anomers.
- (NCERT Ed. 2025-26 12th Part-II Page No. 286)

81. (1)
- meq. of H_2SO_4 = meq. of NH_3 + meq. of NaOH
- $$2 \times \frac{1}{10} \times 40 = \text{meq. of } \text{NH}_3 + 10 \times 1 \times \frac{1}{10}$$
- meq. of NH_3 = 7
- $$\% \text{ of N} = \frac{\text{meq of } \text{NH}_3 \times 1.4}{\text{mass of compound}} = \frac{7 \times 1.4}{0.5} = 19.6\%$$
- (NCERT Ed. 2025-26 11th Part-II Page No. 288)

82. (2)
- Reactivity order due to steric and electronic effect:
-
- (-M) (-I) (+I)
- (NCERT Ed. 2025-26 12th Part-II Page No. 236)

83. (2)
- Only 1°-aliphatic amines can be prepared by Gabriel phthalimide reaction.
- (NCERT Ed. 2025-26 12th Part-II Page No. 264)

84. (1)
- Glycine : $\text{CH}_2 \begin{matrix} \nearrow \text{NH}_2 \\ \searrow \text{COOH} \end{matrix}$
 - Glutamine is a non-essential amino acid.
- (NCERT Ed. 2025-26 12th Part-II Page No. 290, 291)

85. (4)
- Differential extraction method of purification is based on relative solubility in two different solvents.
- (NCERT Ed. 2025-26 11th Part-II Page No. 281)

86. (1)
- | | |
|---|-----------------|
| $>\text{C}=\text{NR}$ | Schiff's base |
| $>\text{C}=\text{N}-\text{OH}$ | Oxime |
| $>\text{C}=\text{N}-\text{NH}-\text{C}_6\text{H}_5$ | Phenylhydrazone |
| $>\text{C}=\text{N}-\text{NHCONH}_2$ | Semicarbazone |
- (NCERT Ed. 2025-26 12th Part-II Page No. 238)

87. (4)
-
- (NCERT Ed. 2025-26 12th Part-II Page No. 273)

88. (1)
- Sucrose is a non-reducing sugar.
- (NCERT Ed. 2025-26 12th Part-II Page No. 287)

89. (1)
- $\text{Na} + \text{C} + \text{N} \rightarrow \text{NaCN}$
- (Sodium Cyanide)
- (NCERT Ed. 2025-26 11th Part-II Page No. 284)

90. (2)
-
- phthalamide phthalimide
- (NCERT Ed. 2025-26 12th Part-II Page No. 253)

91. (2)

The shown complex is succinate dehydrogenase i.e. complex II of mitochondrial ETS.



(NCERT Ed. 2025-26, 11th, Page No. 160)

92. (3)

The ATP synthase enzyme consists of two parts: one called the CF₀ is embedded in the thylakoid membrane. The other portion is called CF₁ and protrudes on the outer surface of the thylakoid membrane. The break down of the gradient provides enough energy to cause a conformational change in the CF₁ particle of the ATP synthase, which makes the enzyme synthesise several molecules of energy-packed ATP.

(NCERT Ed. 2025-26, 11th, Page No. 161)

93. (2)

Glucose and fructose are phosphorylated to give rise to glucose-6-phosphate. This phosphorylated form of glucose then isomerises to produce fructose-6-phosphate. Fructose-6-phosphate undergoes phosphorylation to form Fructose-1, 6-bisphosphate. The fructose 1, 6-bisphosphate is split into dihydroxyacetone phosphate and 3-phosphoglyceraldehyde (PGAL). In the last step of glycolysis, pyruvate is formed from phosphoenol pyruvate.

(NCERT Ed. 2025-26, 11th, Page No. 156)

94. (2)

Ingenhousz in an elegant experiment with an aquatic plant showed that in bright sunlight, small bubbles were formed around the green parts of the plant. Later he identified these bubbles to be of oxygen. Hence he showed that it is only the green part of the plants that could release oxygen. The use of radioactive ¹⁴C by Calvin in algal photosynthesis studies led to the discovery that the first CO₂ fixation product was a 3-carbon organic acid. It was not until about 1854 that Julius von Sachs provided evidence for production of glucose when plants grow. Glucose is usually stored as starch. His later studies showed that the green substance in plants (chlorophyll as we know it now) is located in special

bodies (later called chloroplasts) within plant cells. Using a prism, T.W. Engelmann split light into its spectral components and then illuminated a green alga, *Cladophora*, placed in a suspension of aerobic bacteria. The bacteria were used to detect the sites of O₂ evolution.

(NCERT Ed. 2025-26, 11th, Page No. 134)

95. (3)

Aerobic respiration is the process that leads to a complete oxidation of organic substances in the presence of oxygen, and releases CO₂, water and a large amount of energy present in the substrate. During it, glycolysis occurs in the cytoplasm and TCA cycle and ETS occur in the mitochondria.

(NCERT Ed. 2025-26, 11th, Page No. 158)

96. (4)

A chromatographic separation of the leaf pigments shows that the colour that we see in leaves is not due to a single pigment but due to four pigments: Chlorophyll *a* (bright or blue green in the chromatogram), chlorophyll *b* (yellow green), xanthophylls (yellow) and carotenoids (yellow to yellow-orange).

(NCERT Ed. 2025-26, 11th, Page No. 137)

97. (2)

Electrons from NADH produced in the mitochondrial matrix during citric acid cycle are oxidised by an NADH dehydrogenase (complex I), and electrons are then transferred to ubiquinone. Ubiquinone also receives reducing equivalents via FADH₂ (complex II) that is generated during oxidation of succinate in the citric acid cycle.

(NCERT Ed. 2025-26, 11th, Page No. 160)

98. (1)

The description of structure and variation of living organisms over a period of time, ended up as two, apparently irreconcilable perspectives on biology. The two perspectives essentially rested on two levels of organisation of life forms and phenomena. One described at organismic and above level of organisation while the second described at cellular and molecular level of organisation. The first resulted in ecology and related disciplines. The second resulted in physiology and biochemistry.

(NCERT Ed. 2025-26, 11th, Page No. 131)

99. (3)

There are three points in the Krebs' cycle where NAD^+ is reduced to $\text{NADH} + \text{H}^+$ and one point where FAD^+ is reduced to FADH_2 .

(NCERT Ed. 2025-26, 11th, Page No. 159)

100. (2)

Though chlorophyll is the major pigment responsible for trapping light, other thylakoid pigments like chlorophyll *b*, xanthophylls and carotenoids, which are called accessory pigments, also absorb light and transfer the energy to the reaction centre chlorophyll *a*.

(NCERT Ed. 2025-26, 11th, Page No. 138)

101. (4)

In both lactic acid and alcohol fermentation not much energy is released; less than seven per cent of the energy in glucose is released and not all of it is trapped as high energy bonds of ATP.

Yeasts poison themselves to death when the concentration of alcohol reaches about 13 per cent.

(NCERT Ed. 2025-26, 11th, Page No. 157)

102. (1)

In C_3 plants, the primary CO_2 acceptor is ribulose biphosphate (RuBP). In C_4 plants, it is phosphoenol pyruvate (PEP).

(NCERT Ed. 2025-26, 11th, Page No. 143 & 146)

103. (3)

It is possible to make calculations of the net gain of ATP for every glucose molecule oxidised; but in reality this can remain only a theoretical exercise. These calculations can be made only on certain assumptions that:

- There is a sequential, orderly pathway functioning, with one substrate forming the next and with glycolysis, TCA cycle and ETS pathway following one after another.
- The NADH synthesised in glycolysis is transferred into the mitochondria and undergoes oxidative phosphorylation.
- None of the intermediates in the pathway are utilised to synthesise any other compound.
- Only glucose is being respired – no other alternative substrates are entering in the pathway at any of the intermediary stages.

(NCERT Ed. 2025-26, 11th, Page No. 161)

104. (3)

When only PS I is functional, the electron is circulated within the photosystem and the phosphorylation occurs due to cyclic flow of electrons. A possible location where this could be happening is in the stroma lamellae. While the membrane or lamellae of the grana have both PS I and PS II the stroma lamellae membranes lack PS II as well as NADP reductase enzyme. The excited electron does not pass on to NADP^+ but is cycled back to the PS I complex through the electron transport chain. The cyclic flow hence, results only in the synthesis of ATP, but not of $\text{NADPH} + \text{H}^+$.

(NCERT Ed. 2025-26, 11th, Page No. 140)

105. (3)

The strategy that the plant cell uses is to catabolise the glucose molecule in such a way that not all the liberated energy goes out as heat. The key is to oxidise glucose not in one step but in several small steps enabling some steps to be just large enough such that the energy released can be coupled to ATP synthesis.

(NCERT Ed. 2025-26, 11th, Page No. 155)

106. (2)

RuBisCO has a much greater affinity for CO_2 when the CO_2 : O_2 is nearly equal. This binding is competitive. It is the relative concentration of O_2 and CO_2 that determines which of the two will bind to the enzyme. RuBisCO also catalyses a wasteful oxygenation reaction in C_3 plants: photorespiration. In C_3 plants some O_2 does bind to RuBisCO, and hence CO_2 fixation is decreased. Here the RuBP instead of being converted to 2 molecules of PGA binds with O_2 to form one molecule of phosphoglycerate and phosphoglycolate (2 Carbon) in a pathway called photorespiration. In the photorespiratory pathway, there is neither synthesis of sugars, nor of ATP. Rather it results in the release of CO_2 with the utilisation of ATP. In the photorespiratory pathway there is no synthesis of ATP or NADPH. The biological function of photorespiration is not known yet. In C_4 plants photorespiration does not occur. Now that you know that the C_4 plants lack photorespiration, you probably can understand why productivity and yields are better in these plants as compared to C_3 plants.

(NCERT Ed. 2025-26, 11th, Page No. 147)

107. (1)

2-C compounds: Phosphoglycolate and Acetyl CoA

3-C compound: Phosphoglycerate

4-C compounds: Succinate, Aspartic acid

5-C compound: α -ketoglutarate

Tripalmitin: $C_{51}H_{98}O_6$

(NCERT Ed. 2025-26, 11th, Page No. 147 & 159)

108. (1)

Plants, unlike animals, have no specialised organs for gaseous exchange but they have stomata and lenticels for this purpose. There are several reasons why plants can get along without respiratory organs. First, each plant part takes care of its own gas-exchange needs. There is very little transport of gases from one plant part to another. Each living cell in a plant is located quite close to the surface of the plant.

(NCERT Ed. 2025-26, 11th, Page No. 154)r

109. (2)

Like in respiration, in photosynthesis too, ATP synthesis is linked to development of a proton gradient across a membrane. This time these are the membranes of thylakoid. There is one difference though, here the proton accumulation is towards the inside of the membrane, i.e., in the lumen. Thus, the pH of the lumen decreases. The atmospheric CO_2 is reduced to sugar in Calvin cycle.

(NCERT Ed. 2025-26, 11th, Page No. 141)

110. (3)

Because the respiratory pathway is involved in both anabolism and catabolism, it would hence be better to consider the respiratory pathway as an amphibolic pathway rather than as a catabolic one.

(NCERT Ed. 2025-26, 11th, Page No. 162)

111. (1)

Glycolysis is the breakdown of one molecule of glucose into two molecules of pyruvate. During this, a total of 4 ATP molecules are produced via substrate-level phosphorylation. One turn of the Krebs cycle processes one molecule of Acetyl-CoA. During this cycle, two decarboxylation reactions occur: one when isocitrate is converted to α -ketoglutarate, and a second when α -ketoglutarate is converted to succinyl-CoA. This means that for each Acetyl-CoA that enters the cycle, 2 molecules of CO_2 are released. One turn of the Calvin cycle

refers to the fixation of one molecule of CO_2 . To reduce the two molecules of 3-PGA that are formed from this fixation, 2 molecules of NADPH are utilized. Non-cyclic photophosphorylation is initiated by the photolysis (splitting) of water at Photosystem II to provide electrons for the electron transport chain. The splitting of one molecule of H_2O releases 2 electrons, 2 protons (H^+), and half a molecule of O_2 . This water molecule is utilized in the process.

(NCERT Ed. 2025-26, 11th, Page No. 156, 159, 145, 139)

112. (3)

Regeneration of the CO_2 acceptor molecule RuBP is crucial if the Calvin cycle is to continue uninterrupted.

(NCERT Ed. 2025-26, 11th, Page No. 144)

113. (1)

Citrate is isomerised to isocitrate. It is followed by two successive steps of decarboxylation, leading to the formation of α -ketoglutaric acid and then succinyl-CoA. In the remaining steps of citric acid cycle, succinyl-CoA is oxidised to OAA (via fumarate and malate) allowing the cycle to continue.

(NCERT Ed. 2025-26, 11th, Page No. 159)

114. (4)

C_4 plants show CO_2 saturation at $360 \mu L^{-1}$. The primary CO_2 acceptor is a 3-carbon molecule phosphoenol pyruvate (PEP) and is present in the mesophyll cells. The enzyme responsible for this fixation is PEP carboxylase or PEPcase. It is important to register that the mesophyll cells lack RuBisCO enzyme. The C_4 acid OAA is formed in the mesophyll cells.

(NCERT Ed. 2025-26, 11th, Page No. 145 & 150)

115. (1)

ATP is utilised at two steps in glycolysis: first in the conversion of glucose into glucose 6-phosphate and second in the conversion of fructose 6-phosphate to fructose 1, 6-bisphosphate.

(NCERT Ed. 2025-26, 11th, Page No. 156)

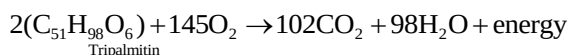
116. (2)

Green plants carry out 'photosynthesis', a physico-chemical process by which they use light energy to drive the synthesis of organic compounds. Ultimately, all living forms on earth depend on sunlight for energy.

(NCERT Ed. 2025-26, 11th, Page No. 133)

117. (3)

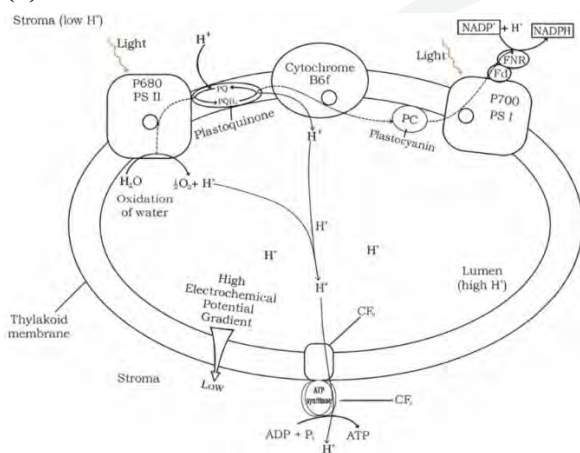
When carbohydrates are used as substrate and are completely oxidised, the RQ will be 1, because equal amounts of CO_2 and O_2 are evolved and consumed, respectively. When proteins are respiratory substrates the ratio would be about 0.9. When fats are used in respiration, the RQ is less than 1. Calculations for a fatty acid, tripalmitin, if used as a substrate is shown:



$$\text{RQ} = \frac{102\text{CO}_2}{145\text{O}_2} = 0.7$$

(NCERT Ed. 2025-26, 11th, Page No. 164)

118. (2)



ATP synthesis through chemiosmosis

(NCERT Ed. 2025-26, 11th, Page No. 141)

119. (3)

Oxidation of one molecule of NADH gives rise to 3 molecules of ATP, while that of one molecule of FADH_2 produces 2 molecules of ATP.

(NCERT Ed. 2025-26, 11th, Page No. 160)

120. (1)

Light reactions or the 'Photochemical' phase include light absorption, water splitting, oxygen release, and the formation of high-energy chemical intermediates, ATP and NADPH. Several protein complexes are involved in the process. The pigments are organised into two discrete photochemical light harvesting complexes (LHC) within the Photosystem I (PS I) and Photosystem II (PS II). These are named in the sequence of their discovery, and not in the sequence in which they

function during the light reaction. The LHC are made up of hundreds of pigment molecules bound to proteins. Each photosystem has all the pigments (except one molecule of chlorophyll *a*) forming a light harvesting system also called antennae. These pigments help to make photosynthesis more efficient by absorbing different wavelengths of light. The single chlorophyll *a* molecule forms the reaction centre.

(NCERT Ed. 2025-26, 11th, Page No. 138)

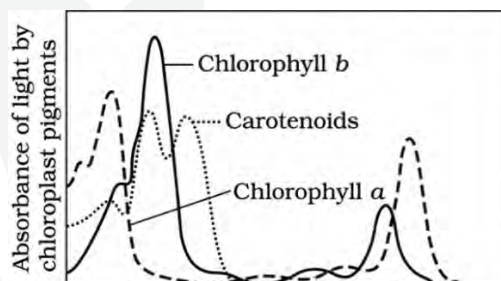
121. (3)

Fermentation takes place under anaerobic conditions in many prokaryotes, unicellular eukaryotes and in germinating seeds. In animal (multicellular) cells also, like muscles during exercise, when oxygen is inadequate for cellular respiration pyruvic acid is reduced to lactic acid by lactate dehydrogenase.

(NCERT Ed. 2025-26, 11th, Page No. 164)

122. (4)

Chlorophyll *b*, xanthophylls and carotenoids are called accessory pigments.



Graph showing the absorption spectrum of chlorophyll *a*, *b* and the carotenoids

(NCERT Ed. 2025-26, 11th, Page No. 137)

123. (3)

Alcoholic fermentation occurs in the cytoplasm.

(NCERT Ed. 2025-26, 11th, Page No. 157 & 158)

124. (4)

There is a linear relationship between incident light and CO_2 fixation rates at low light intensities. At higher light intensities, gradually the rate does not show further increase as other factors become limiting. Light saturation occurs at 10 per cent of the full sunlight.

(NCERT Ed. 2025-26, 11th, Page No. 149)

125. (1)

All carbohydrates are usually first converted into glucose before they are used for respiration. If fatty acids were to be respired they would first be degraded to acetyl CoA and enter the pathway. The proteins would be degraded by proteases and the individual amino acids would enter the pathway at some stage within the Krebs' cycle or even as pyruvate or acetyl CoA. Glycerol would enter the pathway after being converted to PGAL.

(NCERT Ed. 2025-26, 11th, Page No. 163)

126. (3)

The current availability of CO₂ levels is limiting to the C₃ plants. Except for plants in shade or in dense forests, light is rarely a limiting factor in nature. The dark reactions being enzymatic are temperature controlled. The C₄ plants respond to higher temperatures and show higher rate of photosynthesis while C₃ plants have a much lower temperature optimum. Water stress causes the stomata to close hence reducing the CO₂ availability. The dark reactions being enzymatic are temperature controlled. Though the light reactions are also temperature sensitive they are affected to a much lesser extent.

(NCERT Ed. 2025-26, 11th, Page No. 149-150)

127. (2)

Pyruvate, which is formed by the glycolytic catabolism of carbohydrates in the cytosol, after it enters mitochondrial matrix undergoes oxidative decarboxylation by a complex set of reactions catalysed by pyruvic dehydrogenase. The reactions catalysed by pyruvic dehydrogenase require the participation of several coenzymes, including NAD⁺ and Coenzyme A.

(NCERT Ed. 2025-26, 11th, Page No. 158)

128. (3)

The Calvin cycle (biosynthetic process) process does not directly depend on the presence of light but is dependent on the products of the light reaction, i.e., ATP and NADPH. Immediately after light becomes unavailable, the biosynthetic process continues for some time, and then stops. If then, light is made available, the synthesis starts again.

(NCERT Ed. 2025-26, 11th, Page No. 142)

129. (1)

Conversion of Ubiquinol to Ubiquinone: This is an oxidation reaction that occurs in the Electron Transport System (specifically at Complex III). Ubiquinol (the reduced form) donates its electrons

and is oxidized back to Ubiquinone. This process does not create ATP directly; instead, the energy released helps pump protons across the inner mitochondrial membrane. The resulting proton gradient is then used for oxidative phosphorylation. This is not substrate-level phosphorylation.

(NCERT Ed. 2025-26, 11th, Page No. 160)

130. (2)

The products of light reaction are ATP, NADPH and O₂. Of these O₂ diffuses out of the chloroplast while ATP and NADPH are used to drive the processes leading to the synthesis of food, more accurately, sugars. CO₂ is reduced to synthesise glucose during dark reactions.

(NCERT Ed. 2025-26, 11th, Page No. 142)

131. (4)

For each ATP produced, 4 H⁺ pass through F₀ from the intermembrane space to the mitochondrial matrix down the electrochemical proton gradient.

(NCERT Ed. 2025-26, 11th, Page No. 161)

132. (2)

From Glycolysis: 2 NADH + H⁺

From Pyruvate Oxidation: 2 NADH + H⁺

From Krebs' Cycle: 6 NADH + H⁺

Total = 10 NADH + H⁺

(NCERT Ed. 2025-26, 11th, Page No. 156-159)

133. (3)

RuBisCO that is the most abundant enzyme in the world.

(NCERT Ed. 2025-26, 11th, Page No. 147)

134. (4)

Complex IV refers to cytochrome *c* oxidase complex containing cytochromes *a* and *a*₃, and two copper centres.

(NCERT Ed. 2025-26, 11th, Page No. 160)

135. (1)

The scheme of glycolysis was given by Gustav Embden, Otto Meyerhof, and J. Parnas, and is often referred to as the EMP pathway. Calvin along with J.A. Bassham studied reactions in green plants forming sugar and other substances from raw materials like carbon dioxide, water and minerals by labelling the carbon dioxide with C¹⁴.

(NCERT Ed. 2025-26, 11th, Page No. 155)

136. (2)

Pelvic Inflammatory Disease (PID) caused by *Chlamydia* often results in inflammation of the fallopian tubes. If treatment is delayed, the inflammation leads to scarring and fibrosis, which causes tubal blockage. Blocked fallopian tubes prevent the sperm from meeting the egg, leading to infertility. Since natural conception is difficult with blocked tubes, Assisted Reproductive Technologies (ART) are often required.

(NCERT Ed. 2025-26, 12th, Page No. 47)

137. (3)

- IVF (In Vitro Fertilisation) is defined as fertilization outside the body in culture conditions that simulate the body's environment, followed by embryo transfer (ET).
- ZIFT (Zygote Intrafallopian Transfer) involves transferring the zygote or early embryo with up to 8 blastomeres into the fallopian tube. If the embryo has more than 8 blastomeres (e.g., 32), it is transferred into the uterus (Intra Uterine Transfer - IUT), not the fallopian tube.
- GIFT (Gamete Intrafallopian Transfer) involves transferring an unfertilized ovum (gamete) collected from a donor into the fallopian tube of another female who cannot produce one.
- ICSI (Intracytoplasmic Sperm Injection) is a specialized procedure often used when the male partner has severe infertility issues, such as a very low sperm count or poor motility.
- AI (Artificial Insemination) or IUI (Intrauterine Insemination) involves placing sperm into the uterus. For fertilization to occur, the sperm must swim to the fallopian tube to meet the egg. If the tubes are completely blocked, the sperm cannot reach the egg; therefore, AI is ineffective/cannot be used to achieve pregnancy.

(NCERT Ed. 2025-26, 12th, Page No. 48)

138. (2)

List-I (Type of natural selection)	List-II (Characteristics)
Directional selection	Selection pressure favours one extreme phenotype
Stabilizing selection	Selection favoring intermediate phenotypes
Disruptive selection	Selection pressure favours extreme phenotypes at both ends
Artificial selection	Human breeding for specific traits

(NCERT Ed. 2025-26, 12th, Page No. 121)

139. (1)

The most appropriate contraceptive for this couple is copper T (IUD). This is because the female partner suffers from severe nausea with oral pills, making hormonal methods (like oral pills or hormone-releasing IUDs) unsuitable due to potential side effects. The male partner has objections to sterilization, ruling out vasectomy. The Copper T is a non-hormonal, long-acting, and fully reversible device that prevents pregnancy without the systemic side effects associated with hormones or the permanence of surgery.

(NCERT Ed. 2025-26, 12th, Page No. 44)

140. (3)

Gene flow involves the migration of individuals between populations, which mixes the gene pools and reduces genetic differences, thereby equalizing allele frequencies. Genetic drift is stronger in small populations; Mutation is random and directionless; Natural selection is a non-random process.

(NCERT Ed. 2025-26, 12th, Page No. 122)

141. (4)

- Counselling helps prevent unwanted pregnancies, thereby reducing the need for illegal abortions.
- Contraceptives are not a regular requirement for maintaining reproductive health; they are used to prevent natural reproductive events (pregnancy).
- In vasectomy, the incision is made on the scrotum, not the penis.
- Periodic abstinence is generally considered to have low effectiveness (high failure rate), though its reversibility is excellent. The statement claims it is "highly effective" but has "poor reversibility," which is the opposite of reality.

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142. (1)

- Louis Pasteur demonstrated Biogenesis (life comes from life).
- Oparin was from Russia and Haldane was from England. The statement incorrectly pairs Oparin with England and Haldane with Russia.

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143. (1)

- Adaptations increase fitness.
- Variations arise from both mutations and genetic recombination (during sexual reproduction).
- Darwinian theory focuses on natural selection and differential reproduction.
- Lamarck proposed the theory of inheritance of acquired characteristics.
- Genetic drift acts most significantly in small populations.

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144. (2)

The presence of gill slits (pharyngeal pouches) in human embryos is evidence of evolution (recapitulation), suggesting that humans share a common ancestor with fish and other aquatic vertebrates.

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145. (1)

The Lactational Amenorrhea Method (LAM) works because intense breastfeeding leads to high levels of prolactin, which inhibits the release of gonadotropins (GnRH, LH, FSH). This suppression prevents ovulation (Reason). Since ovulation does not occur, conception cannot happen, making the method effective for up to 6 months postpartum.

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146. (4)

Genetic drift refers to the random change in allele frequencies across generations. This effect is mathematically most significant in small populations, where chance events (like the accidental death of a few individuals) can drastically alter the gene pool.

Large populations buffer against random changes; natural selection is a non-random directional force; and high gene flow tends to stabilize allele frequencies rather than causing random fluctuations.

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147. (2)

Hormone-releasing IUDs (like LNG-20 or Progestasert) release small amounts of progestogens. This hormone acts primarily by: Thickening the cervical mucus, which blocks sperm entry (making the cervix hostile).

Changing the lining of the uterus (endometrium), making it unsuitable for the implantation of a fertilized egg.

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148. (3)

List-I	List-II
IVF-ET	Embryo transferred into uterus after <i>in vitro</i> fertilization
ZIFT	Embryo (up to 8 blastomeres) transferred into fallopian tube
GIFT	Unfertilized ovum transferred into fallopian tube
AI	The semen collected either from the husband or a healthy donor is artificially introduced either into the vagina or into the uterus

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149. (3)

The Hardy-Weinberg principle states that allele frequencies remain constant only in the absence of evolutionary influences. Natural selection is a directional force that favors specific alleles (increasing their frequency) and selects against others, thereby disrupting the equilibrium.

Random mating, large population size, and lack of gene flow are conditions that support the equilibrium.

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150. (2)

- Untreated STIs like Chlamydia and Gonorrhea often lead to Pelvic Inflammatory Disease (PID), which causes tubal blockage and infertility.
- STIs like HIV and Hepatitis B can be transmitted via non-sexual routes (blood transfusion, shared needles, mother-to-child).
- Early symptoms are often mild (itching, slight pain) or absent, leading to delayed diagnosis.
- Females are frequently asymptomatic, which facilitates the spread of the infection and complications like PID.
- Hepatitis B and HIV are blood-borne pathogens.

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151. (3)

This is the definition of the Hardy-Weinberg equilibrium ($p + q = 1$; $p^2 + 2pq + q^2 = 1$). In the absence of disturbing factors (mutation, gene flow, selection, genetic drift), the gene pool remains static.

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152. (2)

- Natural selection does not select all variations. It acts as a sieve, selecting only those beneficial variations that increase fitness. Harmful or neutral variations may be eliminated or drift.
- This is the correct definition of natural selection; it acts on heritable traits that confer a reproductive advantage (fitness).

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153. (2)

Under the Medical Termination of Pregnancy (Amendment) Act, 2017, the upper limit for MTP was raised from 20 weeks to 24 weeks for special categories of women (e.g., survivors of sexual assault, minors).A

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154. (2)

Concept that organisms with beneficial traits have increased chances of survival and reproduction is natural selection. This process, described by Charles Darwin, ensures that individuals better adapted to their environment survive and produce more offspring (Differential Reproduction).

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155. (3)

- While early mammals (prototherians) were oviparous, the evolutionary success and better protection of young in mammals is attributed to viviparity (giving birth to live young) and internal development, not oviparity.
- Mammals did not outcompete reptiles "immediately." They co-existed for millions of years (during the Mesozoic era) and only became dominant after the mass extinction of dinosaurs.

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156. (2)

A population pyramid with a broad base and tapering top (triangular) indicates a high birth rate (adding many young people) and a historically short life span (fewer people surviving to old age). While life expectancy is improving, the massive young population is primarily a result of high birth rates in previous decades.

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157. (1)

- Experiments like Joshua Lederberg's replica plating demonstrated that mutations are pre-existing variations. Antibiotics/stressors act as selecting agents, not inducers.
- Hugo de Vries proposed the Mutation Theory based on his observations of the evening primrose (*Oenothera lamarckiana*), distinguishing discontinuous variations (mutations) from Darwin's continuous variations.

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158. (4)

List-I	List-II
Mutation	Source of new genetic variation
Gene flow	Movement of genes between populations
Genetic drift	Random change in allele frequency in small populations
Natural selection	Non-random survival and reproduction based on fitness

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159. (2)

- To tackle the problem of population explosion, the government raised the statutory marriageable age to 18 years for females and 21 years for males. Delaying marriage helps in reducing the reproductive span and total fertility rate.
- The slogan 'Hum Do Hamare Do' translates to "We two, our two." It promotes a two-child norm, not a one-child norm. (Note: NCERT mentions that some young urban working couples have voluntarily adopted a 'one child norm', but the slogan itself is for two).
- The RCH (Reproductive and Child Health) programme was launched in 1997. It is broader than previous family planning programmes because it deals with prevention of sex abuse, sex-related crimes, STDs, postnatal care, and healthy social practices, in addition to contraception.
- 'Hum Do Hamare Do' encourages families to limit their size to two children.
- Incentives were given to couples with small families. It is one of the measures taken to tackle the population problem, implying it was a strategic step, not an ineffective or discontinued one.

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160. (3)

List-I	List-II
HIV	Causes most fatal STI
Genital herpes	Caused by herpes simplex virus
Hepatitis-B	Caused by Hepatitis-B Virus
Genital warts	Caused by Human Papillomavirus (HPV)

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161. (2)

- Thomas Malthus wrote the "Essay on the Principle of Population". He worked on population dynamics (geometric growth of population vs. arithmetic growth of resources), not on "natural selection". Darwin was influenced by Malthus's work on populations and applied those concepts to formulate his own theory of Natural Selection.

- Jean-Baptiste Lamarck, a French naturalist, proposed the theory of evolution known as "Lamarckism". His theory was indeed based on the mechanism of the inheritance of acquired characters driven by the use and disuse of organs (e.g., the elongation of a giraffe's neck).
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162. (4)

The given diagram represents female condom.



Condom for female

- Female condoms are disposable (single-use) devices, just like male condoms. They are not reusable. In contrast, barrier methods like diaphragms, cervical caps, and vaults are reusable.
- They are made of thin sheaths (polyurethane or nitrile, often referred to as rubber/latex. One end of the female condom (the closed inner ring) covers the cervix while the sheath lines the vagina. As a barrier method, they prevent fluid contact and thus protect against Sexually Transmitted Diseases (STDs).

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163. (3)

List-I	List-II
Saheli	Oral contraceptive pill
Vaults	Barrier method
Multiload-375	Intra uterine device
Norplant	Contraceptive implant

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164. (3)

In any large weed population, chance mutations and recombination can produce rare individuals that are naturally resistant to a given herbicide even before its use. When the farmer repeatedly sprays that herbicide, susceptible weeds die, whereas resistant variants survive and reproduce, so the resistant phenotype becomes common in a few seasons; this is natural selection driven by anthropogenic (human) action. This process does not "direct" new resistance mutations in response to need; rather, it sorts pre-existing random variant and showing that evolution is opportunistic rather than purposeful.

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165. (2)

The population explosion was primarily caused by improved health facilities and living conditions, which led to a rapid decline in the Death Rate, Maternal Mortality Rate (MMR), and Infant Mortality Rate (IMR). Simultaneously, the birth rate did not drop as significantly, and there was a large increase in the number of people in the reproducible age group.

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166. (3)

Industrial melanism refers to the evolutionary increase in dark-coloured individuals in polluted, soot-blackened environments, as seen classically in the peppered moth. Here, darker forms are favoured because they are better camouflaged against the dark background, so predators are less likely to see and eat them; thus, it is an example of protective resemblance (camouflage) with the surroundings, not drug resistance, simple smoke-induced darkening, or protection from UV radiation.

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167. (4)

A large number of couples all over the world including India are infertile, i.e., they are unable to produce children inspite of unprotected sexual cohabitation. The reasons for this could be many—physical, congenital, diseases, drugs, immunological or even psychological.

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168. (4)

All animals listed in option (4) are Australian marsupials (Banded anteater is the Numbat).

- Option (1) is incorrect because the Lemur is a placental mammal.
- Option (2) is incorrect because the Flying Squirrel is a placental mammal (the Sugar Glider is the marsupial parallel).
- Option (3) is incorrect because the Bobcat is a placental mammal.

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169. (1)

The world population rose from about 2 billion in 1900 to about 6 billion by 2000 and 7.2 billion in 2011, while India's population rose from about 350 million around independence to nearly 1 billion by 2000 and about 1.2 billion by 2011.

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170. (2)

- While HIV and Hepatitis-B are incurable, Genital Herpes is also not completely curable. The statement implies these are the only two.
- Bacterial and protozoan STIs like Trichomoniasis, Chlamydiasis, Gonorrhoea, and Syphilis are curable if treated early.

- Social stigma is a major barrier to diagnosis.
- The statement "All STIs may lead to... ectopic pregnancies" is too broad. Viral STIs like Herpes or HIV do not typically cause tubal blockages leading to ectopic pregnancy (which is caused by bacterial PID).
- The age group 15–24 years is biologically and socially the most vulnerable.

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171. (1)

- Branching descent and natural selection are the two key concepts of Darwinism.
- Darwin believed that fitness is the end result of the ability to adapt, and this adaptive ability is inherited. Fitness is based on heritable characteristics.

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172. (4)

A deviation indicates that the population is not in equilibrium; it implies that evolutionary forces (drift, mutation, selection, etc.) are at work.

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173. (3)

- Barrier contraceptive methods have very low failure rates when used properly and along with spermicidal creams.
- While diaphragms do not effectively protect against STIs, only condoms do.

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174. (3)

The theory of Spontaneous Generation (Abiogenesis) stated that life originated from decaying and rotting matter like straw and mud. This was later disproven by Louis Pasteur.

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175. (2)

Ovulation typically occurs on day 14. Because sperm can survive for 3-5 days and the ovum survives for about 24 hours, the risk of conception is highest from day 10 to day 17.

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176. (3)

S.L. Miller's spark-discharge experiment observed the formation of amino acids (glycine, alanine, and aspartic acid) from simple inorganic gases.

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177. (4)

List-I	List-II
Copper-T	Increases phagocytosis of sperms and suppresses their motility
Condoms	Prevent conception by physical barrier
Oral contraceptive pills	Inhibit ovulation and implantation; alter cervical mucus quality
Vasectomy	Blocks gamete transport permanently through surgical intervention

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178. (4)

- Lippes Loop is non-medicated.
- Multiload 375 and Copper T are copper-releasing.
- Progestasert and LNG-20 are hormone-releasing.

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179. (1)

- The universe is estimated to be approximately 13.8 billion years old.
- Hydrogen and helium were among the first elements formed. During Big Bang Nucleosynthesis, as the universe cooled, protons and neutrons combined to form the first atomic nuclei, primarily Hydrogen and Helium.
- Earth formed approximately 4.5 billion years ago.
- The early atmosphere was a "reducing" atmosphere and was rich in gases like methane (CH₄), ammonia (NH₃), water vapor, and carbon dioxide. It specifically lacked free Oxygen O₂.
- UV rays broke water vapor into Hydrogen and Oxygen. The released Oxygen first reacted with methane and ammonia (oxidizing them to CO₂ and water). Only after these reducing gases were depleted could free Oxygen accumulate in the atmosphere and interact with UV radiation to form the Ozone layer.

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180. (1)

Antibiotic resistance appears quickly (Assertion) because when the antibiotic is used, it acts as a selection pressure (Reason/Natural Selection). The bacteria that already possess resistance genes (pre-existing variation) survive and reproduce. Because their life cycle is short (part of A), this selected population becomes dominant very rapidly. The mechanism (Natural Selection favoring the resistant strain) explains the phenomenon (rapid evolution of resistance).

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